# Day 33: Practice Problems Suite (Case-Based Scenarios) (Hinglish)

## Context & Alignment

Yeh practice problems Day 33 ke complete unsupervised customer clustering curriculum par strictly based hain. Teeno case studies real-world industry scenarios par formulated hain.

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## Case Study 1: E-Commerce RFM Pipeline aur Feature Engineering

**Scenario**: Aap ek online subscription platform mein Lead ML Engineer hain. Platform ke database mein 50,000 raw customer records hain jisme attributes hain: `Account_Created_Timestamp`, `Birth_Year`, `Monthly_Streaming_Minutes`, `Device_Types`, aur `Subscription_Status` (`Active_Single`, `Active_Family`, `Paused`, `Cancelled`).

1. Mathematically explain karein ki raw timestamps aur raw birth years ko directly Euclidean distance metric mein pass karne se clustering quality kyun destroy hoti hai:

$$d(u, v) = \sqrt{\sum_{i=1}^{p} (u_i - v_i)^2}$$

2. Production retraining pipeline mein reproducibility maintain karne ke liye reference anchor date kaise decide karni chahiye?
3. `Subscription_Status` ko encode karte samay junior engineer `drop_first=True` lagata hai. Explain karein ki supervised regression mein useful hone ke bawajood unsupervised clustering mein `drop_first=True` lagana kyun galat hai?

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## Case Study 2: Optimal K Selection: WCSS Elbow vs Silhouette Score

**Scenario**: Ek engineering team customer clustering ke liye optimal $K \in [2, 12]$ automate kar rahi hai. Experimentation se yeh metrics prapt hue:

| Clusters ($K$) | WCSS (Inertia) | Silhouette Score |
|---|---|---|
| 2 | 24,500 | 0.38 |
| 3 | 16,800 | 0.42 |
| 4 | 11,200 | 0.45 |
| 5 | 9,900 | 0.41 |
| 6 | 8,900 | 0.39 |
| 7 | 8,100 | 0.36 |
| 8 | 7,400 | 0.34 |

1. WCSS (Inertia) ka mathematical objective formula likhein aur explain karein ki $K \to N$ hone par WCSS monotonically decrease kyun hota hai.
2. `kneed.KneeLocator` algorithm mathematically decreasing convex curve par inflection point kaise calculate karta hai?
3. Upar diye gaye table ke basis par optimal $K$ select karein aur rate of change ($\Delta \text{WCSS}$) aur Silhouette score se justify karein.

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## Case Study 3: Cluster Personas aur Marketing Strategy Optimization

**Scenario**: Agglomerative Clustering ($K = 4$, Ward linkage) lagane ke baad marketing analytics team ko yeh summary table milta hai:

| Cluster | Size ($N$) | Mean Income | Mean Total Spend | Kids at Home | Web Visits/Mo | Campaign Response Rate |
|---|---|---|---|---|---|---|
| **Alpha** | 4,200 | \$38,500 | \$210 | 1.35 | 6.8 | 6.5% |
| **Beta** | 2,100 | \$74,200 | \$1,280 | 0.42 | 3.4 | 17.2% |
| **Gamma** | 1,800 | \$35,800 | \$155 | 1.40 | 7.1 | 13.8% |
| **Delta** | 1,400 | \$71,900 | \$1,210 | 0.38 | 3.6 | 33.5% |

CMO ke paas total $100,000 ka marketing budget hai aur unhe revenue maximize karna hai.

1. Charo clusters ko income, spending, family structure aur digital activity ke basis par specific customer personas assign karein.
2. Cluster Delta ka population Alpha se kaafi kam hone ke bawajood use "Golden Cluster" kyun kaha jata hai? Quantitative ROI justify karein.
3. Charo clusters ke liye personalized marketing strategy aur budget allocation playbook design karein.